

Explainable Road Damage Classification Using CNN: Custom CNN vs MobileNetV2 + Grad-CAM + SHAP



POTHOLE



CRACK



MANHOLE

1 PROJECT OVERVIEW



- Objective: Classify road surface images into damage categories and provide explainable predictions.
- Dataset: 18,674 road images
- Classes: Pothole, Crack, Manhole
- Input Size: 128 × 128 pixels
- Goal: Accurate, interpretable and deployable road damage classification

2 ROAD DAMAGE CLASSES

POTHOLE



Bowl-shaped road depression with shadowed interior



CRACK



Thin elongated fracture with variable orientation



MANHOLE



Circular road cover with surrounding pavement



3 METHODOLOGY PIPELINE

1 Data Collection



Kaggle Road Damage Dataset
18,674 images

2 Preprocessing



Resize 128×128
Normalize (0–1)
RGB Images
Split Data

3 Augmentation



Rotation ±15°
Zoom 20%
Horizontal Flip

4 Model Development



Custom CNN
MobileNetV2 (Transfer Learning)

5 Training & Evaluation



Accuracy
Precision
Recall, F1-score
Confusion Matrix

6 Explainable AI



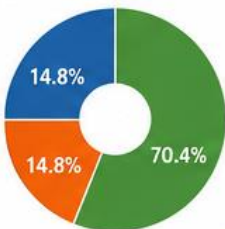
Grad-CAM
Heatmaps
SHAP
Attribution Maps

7 Robustness & Deployment



Blur, Noise,
Brightness Tests
Inference Time
Model Size

4 DATASET & SPLIT

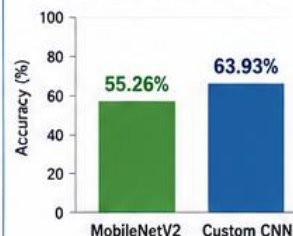


- Training Set
13,148 images (70.4%)
- Validation Set
2,759 images (14.8%)
- Test Set
2,767 images (14.8%)

Total: 18,674 images

5 MODEL COMPARISON (TEST SET RESULTS)

PERFORMANCE COMPARISON



METRICS SUMMARY

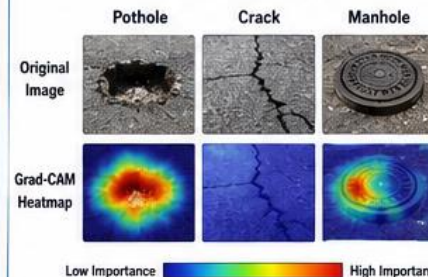
Metric	MobileNetV2	Custom CNN (Our Model)
Test Accuracy	55.26%	63.93%
Test Loss	1.0370	0.8302
Trainable Parameters	—	3,305,027
Model Size (MB)	—	37.87
Inference Time (ms)	—	79.22



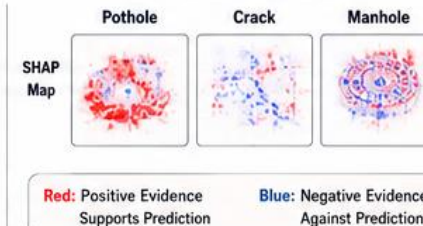
Custom CNN outperforms MobileNetV2 by 8.67% in accuracy.

6 EXPLAINABILITY EXAMPLES (CUSTOM CNN)

GRAD-CAM HEATMAPS



SHAP ATTRIBUTION MAPS



Red: Positive Evidence Supports Prediction
Blue: Negative Evidence Against Prediction

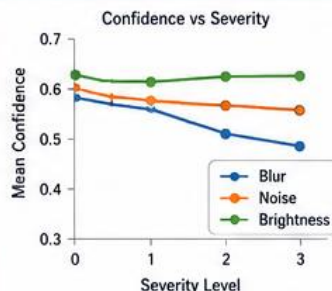
Observations: Model focuses on true damage regions for potholes and manholes, but shows diffuse attention for cracks.

7 ROBUSTNESS ANALYSIS (CUSTOM CNN)

Mean Confidence (Top Prediction)

Degradation	Severity 0 (Original)	Severity 1	Severity 2	Severity 3
Blur (Radius)	0.534	0.514	0.502	0.503
Noise (Std Dev)	0.534	0.528	0.524	0.517
Brightness (Increase)	0.534	0.535	0.550	0.549

- Model shows moderate robustness to blur and noise, and is stable to brightness changes.



8 RESEARCH QUESTIONS



RQ1 Can road images be prepared effectively for three-class classification?



RQ2 Does Custom CNN outperform MobileNetV2 for road damage classification?



RQ3 Which damage class is the most difficult to recognise?



RQ4 Do Grad-CAM and SHAP reveal meaningful regions rather than irrelevant background?



RQ5 Is the model efficient and robust for lightweight deployment?

9 KEY CONTRIBUTIONS & TECH STACK

KEY CONTRIBUTIONS

- Controlled comparison: Custom CNN vs MobileNetV2 on the same dataset
- Dual explainability with Grad-CAM and SHAP for transparent predictions
- Robustness analysis under blur, noise and brightness variations
- Efficiency analysis (inference time, model size) for deployability
- Reproducible pipeline and public code for future research

TECH STACK



FINAL TAKEAWAY: The proposed custom CNN achieves higher accuracy (63.93%), produces interpretable explanations, and is efficient and reasonably robust — making it a strong candidate for real-world road damage screening.

